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IJAZ TAOQEER
03086274142
MUDASSIR NASIR
03197992122

vuhuntpk@gmail.com

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PHY101 FINAL TERM MCQS FILE

1.The prefix mega means

- 10^2
- 10^9
- 10^{-3}
- 10^6

2.Two bodies are falling with the negligible air resistance, side by side, by horizontal plane. If one of the bodies is given an additional horizontal acceleration during its decent it

- Strikes the plane at the same time as the other body
- Strikes the plane earlier than the other body
- Has the vertical component of its velocity altered
- Has the vertical component of its acceleration altered

3.An object moves in a circle at constant speed. The work done by the centripetal force is zero

- The displacement for each revolution is zero
- The average force each revolution is zero
- There is no friction
- The centripetal force is perpendicular to the velocity

4.Two bodies, A and B, have equal kinetic energies. The mass of A is nine times that of B. the ratio of A to that of B is:

- 1:9
- 1:3
- 1:1
- 3:1

5.In simple harmonic motion, the restoring force must be proportional to the;

- Amplitude
- Frequency
- Velocity

- Displacement

6. The second after an electric fan is turned off, the fan rotates at 300 rev/min. its angular acceleration is

- 3.14 rad/s²
- 30 rad/s²
- 30 rev/s²
- 50 rev/mins²

7. The center of gravity coincides with the center of mass

- Always
- Never
- If the center of mass is at the geometrical center of the body
- If the acceleration due to gravity is uniform over the body

8. The ultimate strength of a sample is the stress at which the sample

- Returns to its original shape when the stress is removed
- Remains underwater
- Breaks
- Bends 180

9. The unit of electric field are

- J/m
- J/(C.m)
- J/c
- J.c

10. If 500 J of work is required to carry a charged particle between two points with a potential difference of 20V, the magnitude of the charge on the particle is

- 0.040C
- 12.5C
- 20C
- Cannot be computed unless the path is given\

11. Let Q denote charge, V denote potential difference, and U stored the energy. Of these quantities, capacitors in parallel must have the same

- Q only
- V only
- U only
- Q and U only

12. A magnetic field exerts a force on a charge particle

- Always
- Never
- If the particle is moving across the field lines
- If the particle is moving along the field lines

13. A capacitor in an LC oscillator has a maximum difference of 15V and a maximum energy of 360 μJ . At a certain instant the energy in a capacitor is 40 μJ . At that instant what is the end of introduced of inductor?

- Zero
- 5V
- 10V
- 15V

14. The unit of magnetic dipole moment are

- Ampere
- Ampere. meter
- Ampere. meter²
- Ampere/meter

15. The relation $n_1 \sin \theta_1 = n_2 \sin \theta_2$ is known as _____

- Gauss's law
- Snell's law
- Faraday's law
- Fermat's law

16. Following are the ways by which light can interact with matter, except.

- Emission
- Interference
- Absorption
- Transmission

17. Polarization means _____

- To guide the light only one direction
- To change the frequency of light
- To alter the wavelength of the light
- To separate the light into its colors

18. Wave from two slits are in phase at the slits and travel to a distant screen to produce the third side maximum of the interference pattern. The difference in the distance traveled by the waves is

- Half a wavelength
- A wavelength
- Two wavelengths
- Three wavelengths

19. Which of the following electromagnetic radiations has the photon with the greatest energy?

- Blue light
- Yellow light
- X rays
- Radio waves

20. Which of the following electromagnetic radiations has the photon with the greatest momentum?

- X rays
- Radio waves
- Blue light
- Yellow light

21. A basic postulate of Einstein's theory of relativity is:

- Moving clock run more slowly than when they are rest

- Moving rods are shorter than when they are rest
- Lights have both wave and particle properties
- The law of physics must be the same for observing moving with uniform velocity relative to each other

22.If h is the planck constant, then h is :

- $h/2\pi$
- $h/2\pi$
- h/π
- $2h/\pi$

23.Two events occur simultaneously on the x axis of reference frame s , one at $x = -a$ and the other at $x = +a$. according to an observer moving in the positive x direction

- The event at $x = +a$ occurs first
- The event at $x = -a$ occurs first
- Either event might occur first, depending on the value of a and the observer's speed
- The events are simultaneous

24.How fast should you move away from a 6×10^{14} Hz light source to observe waves with a frequency of 4.0×10^{14} Hz?

- $20c$
- $38c$
- $45c$
- $51c$

25.in Compton scattering from stationary particles the maximum change in the wavelength can be made smaller by using

- less massive particles
- high frequency radiation
- lower frequency radiation
- more massive particles

26.a large collection of nuclei are undergoing alpha decay. The rate of decay at any instant is proportional to:

- the number of undecayed nuclei present at that instant
- the time since the decays started
- the time remaining before all have decayed
- the half life of the decay

27.the inner most part of the sun is known as:

- convection zone
- core
- radiation zone
- nucleus

28.the center of gravity is the average location of the..... of an object

- mass
- weight
- static equilibrium
- dynamic

29.The units of the Planck constant h are those of:

- Energy
- Power
- will not boil
- undergoes some change when heated or cooled

30. Which of the following is true of all virtual images?

- They can be seen but not photographed
- They are ephemeral
- They are smaller than the objects
- E. None of the above

31.In constructing a thermometer it is NECESSARY to use a substance that:

- expands with rising temperature
- expands linearly with rising temperature
- will not freeze
- will not boil

32. A 60-watt light bulb carries a current of 0.5 A. The total charge passing through it in one hour is:

- 120 C
- 3600 C
- $Q/2$,
- $Q, -Q$

33. If the charge on a parallel-plate capacitor is doubled:

- the capacitance is halved
- the capacitance is doubled
- the electric field is halved
- D. the electric field is doubled

34. The wavelength of red light is 700 nm. Its frequency is:

- 4.30×10^4 Hertz
- 4.30×10^5 Hertz
- 4.30×10^3 Hertz
- D. 4.30×10^2 Hertz

35. A capacitor C “has a charge Q ”. The actual charges on its plates are:

- Q, Q
- $Q/2, Q/2$
- $Q, -Q$
- $Q/2, -Q/2$

36. How fast should you move away from a 6.0×10^{14} Hz light source to observe waves with a frequency of 4.0×10^{14} Hz?

- $0.20c$

- $0.45c$
- $0.51c$
- $0.76c$

37.If two light waves are coherent:

- their amplitudes are the same
- their frequencies are the same
- their wavelengths are the same
- D. their phase difference is constant

38.In a photoelectric effect experiment at a frequency above cut off, the stopping potential is proportional to:

- the energy of the least energetic electron before it is ejected
- the energy of the least energetic electron after it is ejected
- the energy of the most energetic electron before it is ejected
- the energy of the most energetic electron after it is ejected

39.The total energy in an LC circuit is 5.0×10^{-6} J. If $L = 25$ mH the maximum current is:

- 10 mA
- 14 mA
- 20 mA
- 28 mA

40.The quantum number m_s is most closely associated with what property of the electron in an atom?

- Magnitude of the orbital angular momentum
- Energy
- z component of the spin angular momentum

- z component of the orbital angular momentum

41.If 500 J of work are required to carry a charged particle between two points with a potential difference of 20 V, the magnitude of the charge on the particle is:

- 0.040 C
- 12.5 C
- 20 C
- cannot be computed unless the path is given

42.The smallest particle of any chemical element that can exist by itself and yet retain the qualities that distinguish it as that element is

- : A. an electron
- B. a proton
- a neutron
- D an atom

43.A particle with a charge of 5.5×10^{-8} C is 3.5 cm from a particle with a charge of -2.3×10^{-8} C. The potential energy of this two-particle system, relative to the potential energy at infinite separation, is:

- 3.2×10^{-4} J
- -3.2×10^{-4} J
- 9.3×10^{-3} J
- -9.3×10^{-3} J
- zero

44. A farad is the same as a:

- J/V
- V/J
- C. C/V
- D. V/C

- E. N/C

45. weber is the same as:

- 1 V/s
- 1 T/s
- 1 T/m

- 1 T · m²

46. coherent sources of light mean

- They must have same speeds
- They must have same frequency
- They must have same amplitude
- They must have fixed phase relative to each other

47. Room temperature is about 20 degrees on the:

- Kelvin scale

- Celsius scale

- Fahrenheit scale
- absolute scale
- C major scale

48. The appearance of color in the thin film is due to:

- Interference

49. When you stand in front of a plane mirror, your image is:

- real, erect, and smaller than you
- real, erect, and the same size as you

- virtual, erect, and smaller than you
- virtual, erect, and the same size as you

50. In a photoelectric effect experiment at a frequency above cut off, the number of electrons ejected is proportional to:

- their kinetic energy
- their potential energy
- the work function
- the frequency of the incident light
- the number of photons that hit the sample

51. The units of magnetic dipole moment are:

- Ampere
- ampere·meter
- ampere·meter²
- ampere/meter

52. The primary of a 3 : 1 step-up transformer is connected to a source and the secondary is connected to a resistor R. The power dissipated by R in this situation is P. If R is connected directly to the source it will dissipate a power of:

- $P/9$
- $P/3$
- P
- 3P

53. We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5 H. We also need a capacitance of

- 1 F
- 1 mF
- C. 1 μ F

- $100\ \mu\text{F}$

54. An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

- a change in entropy
- an increase of pressure
- a change in temperature
- a decrease of internal energy

55. A step-down transformer is used to:

- increase the power
- decrease the power
- increase the voltage
- decrease the voltage

56. The Stern-Gerlach experiment makes use of:

- a strong uniform magnetic field
- a strong non-uniform magnetic field
- a strong uniform electric field
- a strong non-uniform electric field

57. Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in series must have the same:

- Q only
- V only
- U only

- Q and U only

58. "The sum of the currents into a junction equals the sum of the currents out of the junction" is a consequence of:

- Newton's third law

- Ohm's law
- Newton's second law
- conservation of energy

59. The sum of the emf's and potential differences around a closed loop equals zero" is a consequence of:

- Newton's third law
- Ohm's law
- Newton's second law

- conservation of energy

60. Evidence for the wave nature of matter is:

- electron diffraction experiments of Davisson and Germer

- Thompson's measurement of e/m
- Young's double slit experiment
- the Compton effect

61. A conducting sphere with radius R is charged until the magnitude of the electric field just outside its surface is E . The electric potential of the sphere, relative to the potential far away, is:

- zero
- E/R
- E/R^2
- ER

62. An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

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- change in entropy
- an increase of pressure
- a change in temperature
- a decrease of internal energy

63. Capacitors A and B are identical. Capacitor A is charged so it stores 4 J of energy and capacitor B is uncharged. The capacitors are then connected in parallel. The total stored energy in the capacitors is now:

- 16 J
- 8 J
- 4 J
- 2 J

64. Capacitors C1 and C2 are connected in parallel. The equivalent capacitance is given by:

- $C_1 C_2 / (C_1 + C_2)$
- $(C_1 + C_2) / C_1 C_2$
- $1 / (C_1 + C_2)$
- $C_1 + C_2$

65. Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by:

- $C_1 C_2 / (C_1 + C_2)$
- $(C_1 + C_2) / C_1 C_2$
- $1 / (C_1 + C_2)$
- C_1 / C_2

66. During an adiabatic process an object does 100 J of work and its temperature decreases by 5 K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process

is:

- 20 J/K
- 24 J/K
- 5 J/K

15J/

K

67.A virtual image is one:

- toward which light rays converge but do not pass through
- from which light rays diverge but do not pass through
- from which light rays diverge as they pass through
- toward which light rays converge and pass through

68.A generator supplies 100 V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

- 1000 V
- 500 V
- 250 V
- 100 V

69.The rate at which electrical energy is used may be measured in:

- watt/second
- watt·secon

watt

- joule·second

70. In a photoelectric effect experiment at a frequency above cut off, the stopping potential is proportional to:

- the energy of the least energetic electron before it is ejected
- the energy of the least energetic electron after it is ejected
- the energy of the most energetic electron before it is ejected
- the energy of the most energetic electron after it is ejected

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